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Bioinspired cross-aligned multilayered nanocellulose films through shear-induced orientation

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Abstract

This study investigates the biomimetic fabrication of multilayered nanocomposite films with a cross-aligned architecture, a design directly inspired by the mechanically robust structure of natural load-bearing materials, such as the intervertebral disc and wood cell wall structure. The films were fabricated by precise layer-by-layer assembly in which cellulose nanocrystals (CNCs) were oriented by controlled shear, with each successive layer rotated perpendicularly to the previous one. Additionally, UV-assisted photopolymerization rapidly locked the aligned structures, resulting in stable and cohesive films composed of locally anisotropic CNC layers. First, the structural anisotropy was tracked from the liquid suspension phase using small-angle light scattering (SALS) and dichroism to understand the influence of their structure on the anisotropic organization of the resulting films. Second, small- and wide-angle X-ray scattering (SAXS and WAXS) analyses of the UV-cured films provided quantitative nanoscale evidence of the permanent CNC alignment in layers, together with the perpendicular CNC orientation across layers after fabrication. By comparison with unidirectionally aligned materials, the mechanical performance of the resulting architecturally-structured films is remarkable, as they exhibit a 51% increase in compression modulus and a 91% improvement in tensile ductility. These findings open up new perspectives for the design of bioinspired materials with tailored structural and mechanical properties, able to mimic natural hierarchical architectures.

1. Introduction

For millions of years, nature has been the master materials engineer, developing systems that integrate strength, toughness, and adaptability through multiscale organization [1]. Biological materials such as human living tissues (*e.g.*, bone [2], intervertebral disc [3,4], arterial wall [5,6], mollusk shells [7] (*e.g.*, nacre [8]), and plant structures (*e.g.*, wood cell wall structures [9]) are complex hierarchical composites. They have outstanding mechanical properties due to the arrangement of structural constituents that are organized from the molecular and nanometric scales to the macroscale to form an architecture that optimizes stress distribution and energy dissipation [10]. For instance, the "brick-and-mortar" arrangement of aragonite platelets and organic matrix in nacre results in a remarkable increase in toughness over its mineral constituents [11], or the staggered assembly of collagen and apatite in bone provides a superior balance of stiffness and toughness [12]. The common design feature is the development of anisotropic organizations controlling the alignment of fibers or lamellae at micro or nanometer length scale, improving directional load-bearing and damage tolerance [13].

Translating natural design principles into synthetic materials belongs to the field of biomimetic structural materials, which aims at reproducing hierarchical organizations and achieving superior mechanical performance [14,15]. Despite substantial progress with polymers, ceramics, and their composites, a major challenge persists in bridging the gap between the exceptional mechanical properties of their

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nanoscale building blocks and their performance at the macroscale [16,17]. This involves not only the construction of robust reinforcements at the nanoscale, but also a precise and optimized control over the spatial distribution and orientation of these building blocks within large-scale structural architectures [18].

Cellulose nanocrystals (CNCs) have emerged as promising, sustainable reinforcements for polymer nanocomposites. CNCs are derived from the acid hydrolysis of native cellulose fibers. They are slender and rigid nanorods with outstanding longitudinal mechanical properties (90–150 GPa Young modulus, 10 GPa tensile strength) [19,20]. They are also renewable, biodegradable, and exhibit an anisotropic behavior in the liquid crystalline regime [21]. CNCs are thus ideal candidates as proper reinforcement in engineering anisotropic polymer nanocomposites with high mechanical properties. Recent studies have demonstrated that CNC content, aspect ratio, and orientation drastically alter the mechanical behavior of the nanocomposites, such as stiffness, strength, and toughness, provided a proper tuning of these first-order structural descriptors [19,22,23]. For example, many processing methods are available to induce the alignment of CNCs during nanocomposite processing/forming, such as mechanical shear, magnetic, or electric fields. However, with low to moderate CNC contents, this usually results in a temporary CNC alignment. Indeed, once the aforementioned stimulus is removed, the CNC Brownian motion and relaxation usually randomize the orientation of the particles [24–27]. To preserve the induced architecture after the physical stimulus towards a proper permanent, load-carrying structure, the aligned state must be durably quenched.

Among the available possibilities to quench CNC architectures, UV-assisted photopolymerization is particularly relevant: by quickly entrapping CNCs within the polymer matrix *via* near-instantaneous cure time, this processing route restrains relaxation and re-orientation. Poly (ethylene glycol) diacrylate (PEGDA) is ideal for this approach due to its compatibility with aqueous CNC suspensions, fast UV-reactivity, and capabilities for producing flexible and biocompatible networks [28,29]. Thus, PEGDA-CNC composites have been designed in the form of hydrogels, photonic films, and tissue scaffolds [30–32]. Ge et al., for example, significantly contributed to the development of CNC-poly (PEGDMA) composites with high toughness (31 MJ m^{-3}), stretchability (104%), strength (42 MPa), and solvent resistance, overcoming the brittleness intrinsically associated with CNC structures [33]. However, the focus has typically been limited to uniformly aligned or self-organizing particles in a single layer, rather than capturing controlled orientations among multiple layers. Other studies reported shear or extensional flow to achieve temporary CNC orientation before curing, which resulted in a higher anisotropy in single-layered films [24,34]. However, no studies focused on UV-quenched shear-aligned CNC nanocomposites in order to design and process novel cross-aligned multilayered CNC architectures with improved mechanical properties.

Hence, the present work presents a novel strategy for fabricating bio-inspired multilayered CNC-PEGDA nanocomposites featuring precisely controlled, anisotropic, cross-aligned CNC architectures. Successive layers were formed *via* perpendicular rotation of a substrate. Each layer was composed of shear-induced, UV-quenched, aligned CNCs, mimicking the hierarchical logic of natural load-bearing (nano)fiber-reinforced composites, such as wood cell wall structures. The anisotropy variation of the CNC-PEGDA native suspensions upon shearing was systematically characterized using small-angle light scattering (SALS), dichroism, and small-angle X-ray scattering (SAXS) to resolve the nanoscale orientation of CNCs within the cured layers of the nanomaterials, while microfocus wide-angle X-ray scattering (WAXS) analyses provided complementary data on the CNC orientation at the molecular scale. The mechanical properties of the cross-aligned films were investigated using tensile and compression tests. The results demonstrated our scalable and effective strategy to translate nanoscale anisotropy into macroscale mechanical functionality, based on the perpendicular alignment of CNCs to create sustainable, high-performance composites with tunable anisotropic properties and enhanced structural performance.

2. Materials and methods

2.1. Materials and preparation of CNC suspensions

The CNCs were purchased from the UMaine Development Center (University of Maine, Orono, USA) as a 10.6 wt% aqueous suspension. The rodlike particles decorated with sulfate half-ester groups (SO_4^-) with a sulfur content of 1.1 g per 100 g of CNCs, having Na^+ as counterion [28]. The stock suspension was sonicated with a probe sonifier (Branson Digital Sonifier, Marshall Scientific, Hampton, NH; 250 W output power) using a 30% duty cycle and 0.5 W output control in two steps of 10 min and 5 min ($\approx$15.5 kJ.g^{-1}.L^{-1}). The resulting suspension was diluted with deionized water to prepare CNC dispersions of 5 and 6 wt%, followed by vigorous stirring for 2 h. Transmission electron microscopy (TEM) of the CNCs were recorded from preparations negatively stained with 2% uranyl acetate, with a JEOL JEM-2100 Plus microscope operating at 200 kV. Typical images are shown in Supplementary Fig. S1. The CNC dimensions were measured from 225 particles using the ImageJ software (NIH, USA). The rodlike particles had an average length l = 127.4 ± 34.1 nm, a width d = 10.9 ± 2.6 nm, and thus an aspect ratio l/d = 12.3 ± 4.3 [28]. Poly(ethylene glycol) diacrylate (PEGDA, M_n = 700 g mol^{-1}) and the Irgacure 2959 photoinitiator (I2959) were purchased from Sigma-Aldrich (France). PEGDA was selected for its complete miscibility with water, allowing homogeneous mixing with CNC suspensions and subsequent UV crosslinking [35].

2.2. Preparation of PEGDA/CNC suspensions

To prepare the PEGDA/CNC suspensions, the sonicated CNC dispersions were mixed with PEGDA at a 30/70 weight ratio of (PEGDA/aqueous CNC suspension) under magnetic stirring at room temperature (T = 20 °C). This weight ratio was selected as a balance between minimizing material shrinkage upon drying and maintaining a sufficient CNC content in the aqueous phase. Irgacure 2959 (1 wt% relative to PEGDA) was subsequently added under stirring until complete dissolution. Final CNC concentrations in the PEGDA/CNC suspensions were adjusted to 5 and 6 wt%, while the PEGDA fraction remained fixed at 30 wt%. The corresponding formulations will be referred to as P30C5 and P30C6, where P30 refers to 30 wt% PEGDA and C5 or C6 to the CNC concentration (5 or 6 wt%).

2.3. Analysis of the orientation of CNC and PEGDA/CNC suspensions

2.3.1. Rheometry

Rheological measurements were performed using a shear rate-controlled rheometer (DHR-3, TA Instruments) equipped with a cone-plate geometry (60 mm diameter, 2° angle), under the same conditions as the scattering experiments (see below). All measurements were carried out at a controlled temperature of 20 ± 1 °C. An evaporation-proof system was placed on top of the geometry to minimize solvent loss, ensuring saturation of the sample atmosphere.

2.3.2. SALS under shear-flow

The SALS setup was developed as previously described [26,36], and is shown schematically in Supplementary Fig. S2. It consists of a 2-mW He-Ne laser (λ = 632.8 nm), a Fresnel lens for forward light collection, and an Allied Vision MAKO G-419B POE monochrome camera (CMOS sensor, 2048 × 2048 pixels). Images were acquired with Vimba Matlab, and scattering intensities were processed using the SAXSutilities software [37]. Scattering patterns were continuously recorded with the

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camera system during the experiments. The modulus of the scattering vector was calculated following Eq. (1).

$$q = \left(\frac{4\pi}{\lambda}\right) sin\left(\frac{\theta}{2}\right) \quad (1)$$

where λ is the light wavelength in the sample, and θ is the scattering angle.

Measurements covered a q-range of 2×10^{-4} to 4×10^{-3} nm^{-1}. The shear flow cell was a rectangular quartz glass chamber (H = 4 mm, W = 7.4 mm, L = 100 mm), as shown in Fig. S2. However, in this setup, the channel was oriented vertically, unlike typical horizontal configurations used in microscopy. The shear flow cell was connected to a syringe pump to control the flow rate Q and thereby the shear rate in the channel. The shear rate $\dot{\gamma}$ in the rectangular geometry was calculated following Eq. (2) [38].

$$\dot{\gamma} = \frac{6Q}{WH^2} \quad (2)$$

In the experiments, the laser beam spot size was approximately 250 μm and was directed along the x-axis, perpendicular to the flow cell walls, and focused at the mid-plane of the channel (Fig. S2). The Fresnel lens collected the forward-scattered light, which was then transmitted to the camera for recording. The scattering intensities were subsequently processed to obtain azimuthal averages using the SAXSutilities software. The SALS analysis of suspensions also allowed measuring the pitch of the cholesteric phase. The pitch value, P, was calculated from the q_p position in the SALS pattern using $P = 4\pi/q_p$.

2.3.3. Dichroism

To complement the SALS measurements, dichroism ($\Delta n''$) experiments were performed under the same shear flow conditions and with identical CNC suspensions. This method provides information on particle orientation across multiple length scales, from nano to micro. Initially applied to lyotropic sepiolite suspensions under extensional flow, it revealed the "strong flow-weak flow" phenomenon in 3D systems [39]. More recently, it has also been adapted to CNC suspensions to assess orientation changes at the particle scale using a laboratory setup [40]. Dichroism corresponds to the difference between the principal values of the imaginary part of the refractive index tensor n''_{ij}.

The setup used a He-Ne laser (λ = 632 nm), with a *ca.* 250 μm spot size, and a 0.1 s exposure directed through a straight prism, a linear polarizer set at 45° to a photoelastic modulator (PEM), a quarter-wave plate parallel to the polarizer, the sample cell, and a photodiode detector. The transmitted intensity is expressed as:

$$I = I_{dc}\ \left(I + 2\cos 2\alpha \tan h\delta'' \sin \omega t + 2\sin 2\alpha tanh\delta'' \cos 2\omega t\right) \quad (3)$$

where I_{dc} is the direct transmitted intensity, and α is the average fiber orientation angle relative to an arbitrary axis.

The optical anisotropy δ'' is linked to dichroism by:

$$\delta'' = 2\pi\Delta n'' \frac{e}{\lambda} \quad (4)$$

where $\Delta n''$ is the dichroism, and e is the sample thickness.

The signal was analyzed using two lock-in amplifiers tuned to the frequency ($1/\omega$ = 52 kHz) of the PEM. At each shear flow value, from the signal recorded on the photodiode, the initial direct intensity, I_{dc}, as well $I(\omega)/I_{dc}$ and $I(2\omega)/I_{dc}$ were recorded and extracted by the two lock-in amplifiers locked in to the frequency 1/ω. The signals $I(\omega)/I_{dc}$ and $I(2\omega)/I_{dc}$ are proportional, respectively, to sin(ωt) and cos(2ωt), and were used to determine the dichroism value $\Delta n''$ using Eq. (5) [39]:

$$\Delta n'' = \frac{\lambda.\pi.e}{2} \tanh^{-1}\left(\sqrt{\left[\frac{I(\omega)}{I_{dc}}^2\right] + \left[\frac{I(2\omega)}{I_{dc}}\right]^2}\right) \quad (5)$$

with e the thickness of the sample (nm).

2.4. Fabrication of cross-aligned multilayered films

Multilayered PEGDA/CNC films with cross-aligned CNC structures were prepared using a custom-built flow cell developed at Laboratoire Rhéologie et Procédés (LRP), which consists of a flow cell channel coupled to a rotational support (Figs. 1 and S3). The cell consists of a custom-made Teflon cylinder with a 1 mm-thick channel (50 mm-wide × 100 mm-long) and a central rotating stainless-steel support (circular in shape with a diameter of 50 mm) that can move vertically from the quartz glass window at the top to the piston at the bottom. In front of this rotating stage, a 50 mm quartz window, on the top of the cell, allowed UV light to pass through the channel and cure the suspension at the center. A cross-shaped mask was used to fabricate nanocomposites in the form of a cross (3-mm square gauge).

PEGDA/CNC suspensions (30% PEGDA and CNC concentration of 5 and 6 wt% with photoinitiator (1 wt%) were introduced into the channel using a syringe pump to impose a controlled velocity gradient, and thus the related shear rate $\dot{\gamma}$ at 20 s^{-1}, to promote the CNC alignment along the flow direction. After stabilization, the shear flow was stopped, and the suspension was immediately subjected to UV irradiation (λ = 250–350 nm) for 10 s with a UV lamp (Omnicure S2000, France) to fix by curing the PEDGA a first oriented layer, forming a 1 mm-thick layer. After curing, the piston support was lowered by 1 mm and rotated by 90° from the initial position, enabling deposition of a second PEGDA/CNC layer under shear flow, which was then UV-cured to produce a second layer with a CNC orientation perpendicular to the first one. This procedure was repeated to build additional layers (up to 4 overlapping layers), the support being rotated by 90° between each successive curing step. Other rotation angles could also be applied. A cross-shaped mask enabled the identification of the flow-induced orientation in each successive layer. Once UV-curing was completed, the multilayer samples were removed from the cell and left at room temperature (T = 20 °C) to allow residual water to evaporate for 2 days, forming a cross (14.5 mm × 12.5 mm). A shrinkage was observed, but it did not affect the cross-shaped geometry of the final multilayer films. Films with 2 or 4 cross-aligned layers (P30C5 and P30C6 compositions) were prepared. Uniaxially-oriented films were also prepared, and their mechanical properties were compared with those of the cross-aligned samples.

The dry film resulting from the PEGDA-CNC mixture exhibits a CNC mass fraction of 12.3% for an initial 5 wt% CNC solution and 14.9% for an initial 6 wt% CNC solution. In terms of relative volume, the CNC occupied approximately 7.68 vol% in the film for the 5% initial solution, and 8.75 vol% for the 6% initial solution.

2.5. Characterization of the cross-aligned multilayer nanocomposites

2.5.1. Analysis of the structural organization by small- and wide-angle synchrotron X-ray scattering (SAXS/WAXS)

SAXS and WAXS experiments were carried out at the European Synchrotron Radiation Facility (ESRF, Grenoble). SAXS was performed at the TRUSAXS ID02 beamline, using 12.232 keV photons at room temperature (T = 20 °C). Sample-detector distances of 1.5 and 10 m provided a q-range of 0.02–2 nm^{-1}, where q was defined as in Eq. (1). Thanks to the pinhole collimation setup available at ID02, all the measurements were performed using a beam cross-section (FWHM) at the sample position of about 25 μm vertically and 50 μm horizontally. Films were first positioned on a sample holder with their surface perpendicular to the incident beam (top view observation) as shown in Fig.

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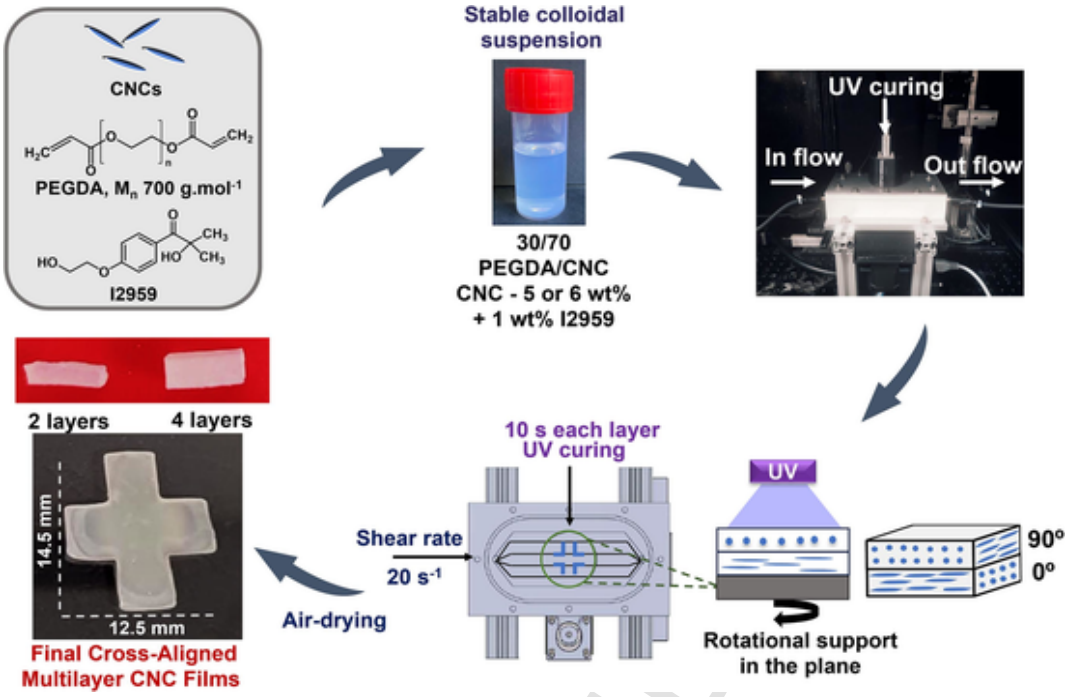


Fig. 1. Schematic illustration of the fabrication process for cross-aligned multilayer CNC films. PEGDA, photoinitiator (I2959), and CNC suspensions were mixed at a 30/70 weight ratio, UV-cured in the cell. After deposition of 2 or 4 layers, the films are removed and air-dried, resulting in final cross-aligned multilayer CNC films (see detailed setup in Supplementary Data**, Fig. S3**).

2a. After the samples were measured along the z-axis (side view), corresponding to the thickness, as shown in Fig. 2b. SAXS patterns were recorded every 100 μm over the entire height of the film. The two-dimensional (2D) scattering intensity patterns were collected using a high-resolution array detector (EIGER2-4 M detector, Dectris) [41].

The analysis of the anisotropy of the intensity/orientation distribution function (ODF) of the patterns was performed using the MATLAB-based Small-Angle Scattering Evaluation Tool (SASET) [42] from 0.20 to 0.70 nm^{-1}. The model-free principal component analysis (PCA) method was applied, yielding anisotropy values from 0 (isotropic) to 1 (fully aligned), quantitatively comparable to the conventional order parameter depending on the system [42]. Annular averages were also calculated using SAXSutilities over a q-range of 0.18 to 0.43 nm^{-1}, to define the scattering intensity as a function of the azimuthal angle (ψ) in the 2D SAXS patterns to quantify both the degree of orientation and its angular variation (Fig. S4a,b).

SAXS-WAXS experiments were performed on the ID13 Microfocus beamline (ESRF, Grenoble), with 13-keV photons at room temperature (T = 20 °C). The sample-to-detector distance of 0.83 m provided a q-range of 0.2–28 nm^{-1}, where q was defined as in Eq. (1). Films were first positioned on a sample holder with their surface perpendicular to the incident beam (top view observation), as shown in Fig. 2a, or side view, as shown in Fig. 2b.

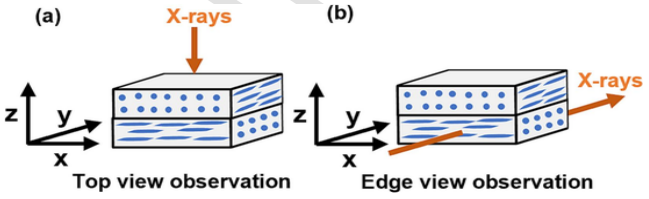


Fig. 2. Schemes of the top (a) and side (b) views that define the orientation of the X-rays mean with respect to the cross-layered samples.

Films were characterized by a SAXS/WAXS mapping using a microfocused X-ray beam (1.8 × 2.7 μm^2) and a 5-μm step to scan 5 lines in the samples. The two-dimensional (2D) scattering intensity patterns were collected using a high-resolution array detector (EIGER2-4M detector, Dectris) [41]. Annular averages were calculated using the SAXSutilities software over two q-ranges, one around the SAXS domain within the 0.3–1.5 nm^{-1} q-range, and the second over the WAXS region with q-vectors ranging from 8 to 27 nm^{-1}. These annular averages were done to define the scattering intensity as a function of the azimuthal scattering angle ψ in the plane of the two-dimensional 2D SAXS patterns to quantify the degree of orientation as well as its angular variation (Fig. S4c). Sector angular averages were also calculated using SAXSutilities around vertical (I_v) (260–280°) and horizontal (I_h) (340–360°) angular sectors, to emphasize the anisotropy of the SAXS/WAXS patterns (Fig. S4d).

To characterize the orientation of the cellulose crystallite axis with respect to the other axes, the degree of orientation (Π) and Hermans' orientation parameter (f) were also calculated from the azimuthal intensity distribution from the most intense peak (200 equatorial reflections) [43,44], Eqs. (6)–(9).

$$\Pi = \frac{180 - \mathrm{FWHM}}{180} \tag{6}$$

$$\phi = \psi_{\mathrm{peak}} - \psi \tag{7}$$

$$f = \frac{3\left\langle \cos^2\phi \right\rangle - 1}{2} \tag{8}$$

$$\left\langle \cos^2\phi \right\rangle = \frac{\sum_0^{\pi/2} I(\phi)\,\sin\phi\cos^2\phi}{\sum_0^{\pi/2} I(\phi)\,\sin\phi} \tag{9}$$

where FWHM corresponds to the full width at half maximum. The parameter ψ is the azimuthal scattering angle in the plane of the two-

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dimensional scattering pattern, while ϕ is the angle between the normals of crystalline planes diffracting and the reference axis ($\phi = \frac{\pi}{2} - \psi$). Thus, $I(\phi)$ represents the intensity as a function of the angle ϕ. The f values range between 0 and 1. $f = 0$ corresponds to isotropic scattering (random nanostructure orientation), and $f = 1$ corresponds to complete alignment of the nanocrystals.

2.5.2. Evaluation of the mechanical properties

The mechanical properties of the nanocomposites were evaluated for the most advanced architectures, consisting of four PEGDA/CNC layers with varying preferred CNC orientations: the CNCs were either aligned in the same direction across all four layers (reference samples), or cross-oriented perpendicularly between two successive layers. The mechanical behavior of all samples was characterized under uniaxial compression and tension.

First, compression tests along the thickness of the produced samples were conducted with a tension-compression electromechanical testing machine (Instron® 5944) equipped with a ± 500 N load cell and with cylindrical compression platens (25 mm in diameter), as shown in Fig. S5a. The samples were tested with an initial thickness of 3.2 mm, a thickness-to-length ratio of 0.4, and a cross-sectional area of 66.4 mm^2. Tests were performed at room temperature ($T = 20$ °C), by compressing samples down to 30% of their initial thickness at 0.1 mm s^{-1}. A small preload of ≈ 0.05 N was applied to align the samples with the compression platens. Force and displacement were recorded to obtain stress-strain curves, while a high-resolution CCD camera (JAI® BM500GE, 15 Hz) captured images to monitor dimensional changes and detect slippage.

Tensile tests along the midplane of samples were performed using a device developed at Laboratoire Rhéologie et Procédés (LRP) [30] equipped with a 100 N load cell (0.1% accuracy), as displayed in Fig. S5b. The samples were tested with an initial gauge length of 20 mm, a length-to-width ratio of 5, and a cross-sectional area of 12.8 mm^2. Tests were performed at room temperature (T = 20 °C), with samples stretched up to rupture at a speed of 0.1 mm s^{-1}. The tensile force F and jaw displacement Δl were recorded to estimate the nominal tensile stress $\sigma = F/S_0$ and strain $\varepsilon = \Delta l/l_0$ from the knowledge of the initial sample gauge length l_0 and cross-section S_0.

The obtained stress-strain data were quantified by several mechanical descriptors: the apparent modulus E, the peak stress achieved during the loading phase, σ_{max} and the strain at break ε_b.

3. Results and discussion

3.1. Shear flow-induced CNC alignment in PEGDA/CNC suspensions

3.1.1. Rheological behavior in steady-state conditions

The steady-state shear viscosity as a function of the applied shear rate was quantified for suspensions containing 5 and 6 wt% CNCs in the absence or presence of PEGDA (P30C5 and P30C6 samples, respectively) (Fig. 3). First, this graph shows that the incorporation of 30 wt% PEGDA resulted in a substantial increase in the shear viscosity, nearly by an order of magnitude. This enhancement was attributed to complex effects involving chiral interactions within the cholesteric CNC structure and the inherently higher viscosity of PEGDA (*ca.* 100 mPa.s) [45].

In addition, whatever the considered suspension, the shear viscosity was non-constant and a complex shear thinning function of the shear rate, which was partly/mainly due to the CNC structuration. Indeed, CNC suspensions are prone to exhibiting three-regime flow behaviors, which are characteristic of liquid crystalline phases [46–48]. At low shear rates, Regime I corresponds to an initial shear-thinning regime in a power-law. A rapid decrease in viscosity is observed, which is attributed to the gradual orientation of cholesteric CNC domains along the flow direction, which accomplishes internal stress relief and improves flow [49]. While increasing the shear rate, the system switches into

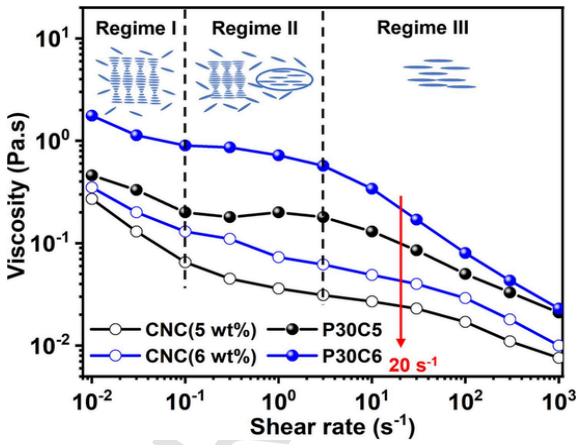


Fig. 3. Steady-state shear viscosity as a function of shear rate for 5 and 6 wt% CNC suspensions and of P30C5 and P30C6 at $T = 20$ °C, showing three distinct regimes.

Regime II with a viscosity plateau that is often referred to as the Newtonian region. Within this regime, CNCs are aligned in a steady-state orientation, and tactoids retain their orientation during moderate shear. There is no further thinning, and a stable liquid crystalline structure is maintained [50,51]. Above this region, the system enters Regime III, with a secondary shear-thinning response: cholesteric domains progressively undergo fragmentation towards a nematic-like arrangement of CNCs. The alignment of CNC reduces internal friction, and the shear viscosity continues to decrease as the shear rates are increased [52,53].

For the fabrication of multilayer CNC films with cross-aligned architectures, we found that a sufficiently high shear rate of about 20 s^{-1} provided an optimal balance, *i.e.*, by inducing controlled CNC alignment along the flow direction while preserving a uniform nematic order. This condition ensured a reliable processing and reproducible anisotropy in the final composites. To directly correlate rheological features with the microstructural evolution, the same range of shear rates was imposed in a channel-type shear flow cell during *in-situ* SALS experiments. The volumetric flow rate Q applied by the syringe pump in each experiment was calculated according to Eq. (2), ensuring that the imposed shear rate $\dot{\gamma}$ precisely matched the values established during conventional rheometric tests.

3.1.2. CNC organization under shear-flow: in-situ SALS

With the help of the different flow regimes previously identified, the dynamic structural evolution of the suspensions at the micrometer scale was probed by SALS under flow using the custom-designed flow cell. The direct SALS observation of structural reorganization and orientation induced by shear was analyzed, as illustrated in Fig. 4 for the sample P30C6. In the shear thinning regime I, *e.g.,* at an applied shear rate of 0.027 s^{-1} (Fig. 4a), the *in-situ* SALS scattering patterns were anisotropic, with a marked peak intensity oriented along the vertical direction, consistent with the occurrence of aligned cholesteric CNC domains, with the helical axes oriented in the vertical direction (perpendicular to the velocity direction). The green region in SALS patterns, Fig. 4a, represents the oriented cholesteric phase, Fig. S6. This orientation facilitated a partial stress release within the suspension, leading to the viscosity decrease as a function of shear rate characteristic of the first shear-thinning regime, illustrated in Fig. 3. The corresponding time-resolved scattering patterns reveal a persistent correlation peak at $q_p = 2.6 \times 10^{-3}$ nm^{-1} with a pitch value $P = 4830$ nm, consistent with the cholesteric pitch value of the CNC suspension. This indicates that the helical periodicity of the cholesteric structure and pitch distance under low shear were presumably preserved. These findings obtained for the first time on PEGDA/CNC suspensions indicate that the

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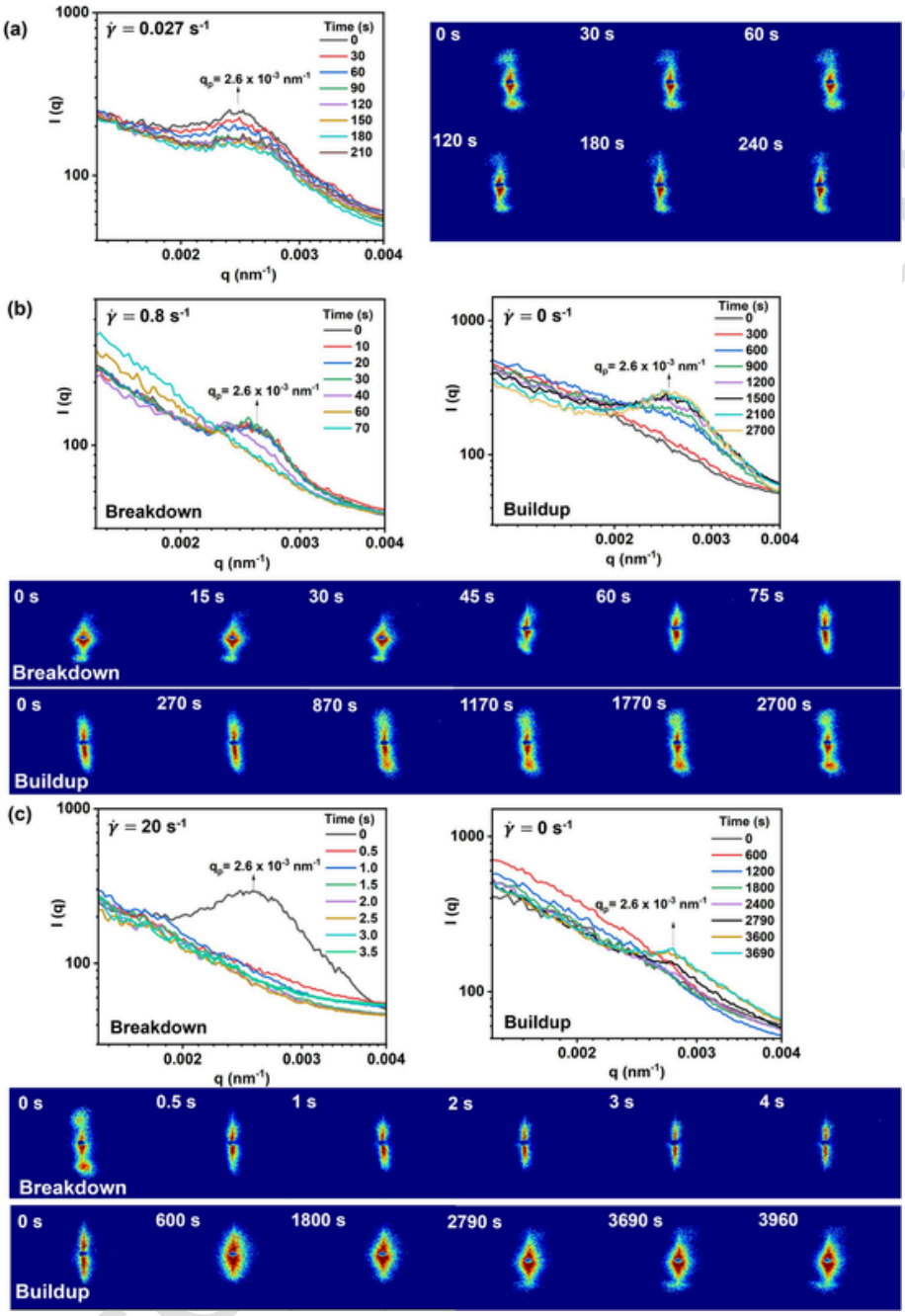


Fig. 4. Transient state start flows of the P30C6 suspensions under *in-situ* SALS. Scattering patterns and averaged intensities as a function of time, with shear rates corresponding to Regimes I (a), II (b), and III (c).

cholesteric phase remained structurally intact during Regime I, in agreement with earlier results reported by Pignon et al. on CNC suspensions in aqueous medium [26].

In Regime II, at a shear rate of 0.8 s^{-1} (Fig. 4b), the cholesteric arrangement of CNCs initially exhibited vertically aligned helical axes. The characteristic correlation peak at $q = 2.6 \times 10^{-3}$ nm^{-1} progressively weakened and disappeared after 60 s. Within this regime, the disappearance of the pitch peak corresponds to a progressive disruption of tactoids into substructures smaller than the pitch distance, making the periodicity undetectable [26]. The same behavior is observed in aqueous CNC suspensions. In contrast, for the latter condition, the same disruption occurred much earlier, at $t = 10.5$ s. This marked difference indicates that the stabilization imparted by PEGDA should originate from the interplay between reduced chiral interactions in the cholesteric structure and the intrinsic viscosity of PEGDA. This finding provides a refined interpretation of the viscosity plateau regime in the range of shear rate, which has traditionally been attributed to the alignment of cholesteric domains along the shear direction, followed by their eventual breakdown. After $t = 75$ s, the shear flow was stopped, and the SALS patterns were recorded as a function of time. At longer relaxation times, a structural recovery was observed, with the reemergence of the cholesteric arrangement after approximately 870 s. Due to the viscosity effect that reduces the mobility of the CNCs in the PEGDA/CNC suspension, the buildup of the cholesteric phase takes a much longer time than its breakdown.

At the beginning of Regime III, for a shear rate of 20 s^{-1} (Fig. 4c), the vertically aligned cholesteric arrangement was rapidly disrupted. Nevertheless, anisotropic SALS patterns with a vertical orientation persisted as a function of time. This would be attributed to a nematic-like orientation of CNCs that persisted up to length scales of about a few micrometers. This effect of stabilization of the nematic structure at the beginning of Regime III could be attributed to the effect of the intrinsic viscosity of PEGDA, which stabilizes the flow around isolated CNCs.

After a shear rate of 20 s^{-1}, the shear was stopped, and the SALS patterns were recorded as a function of time. When the shear was stopped, it took 600 s to start recovering from a nematic domain from nanometer to micrometer scales in a slow relaxation process. This suggests that CNCs rapidly changed their orientation under shear but required a lower build-up of their hierarchical organization once the flow was stopped in the presence of PEGDA. In addition, the increase in viscosity and polymer-mediated interactions should impede particle motion, while the formation of the PEGDA network should restrain mobility within the colloid. Thus, the molecular reorganization of the cholesteric organization was slowed down, leading to the observed slower structural equilibration.

At 2800 s, the onset of the cholesteric correlation peak reappearance was also associated with the pitch of the cholesteric domains in the sample. This is consistent with the reorganization of cholesteric tactoids with a vertical orientation of their helical axis. The delayed recovery was again attributed to the reduced mobility of chiral interactions in the cholesteric structure in the highly viscous PEGDA-rich medium, which delayed the re-establishment of a long-range cholesteric order. In comparison, Pignon et al. [26] reported that aqueous CNC suspensions exhibit recovery of the micrometer-scale cholesteric organization within only a few seconds. In such systems, the constant interparticle distance and uniform interparticle interactions favor rapid reorganization of the particles. Importantly, the slower relaxation observed in PEGDA-containing systems remained compatible with processing requirements such as UV-curing, where flow cessation followed by gradual structural recovery enables a controlled film formation.

3.1.3. CNC organization under shear-flow: dichroism measurements

Experiments similar to those conducted with SALS were performed using dichroism, which provides a complementary probe of CNC orientation under shear flow. The transmitted intensity I_{dc} and dichroism were recorded at the center of the shear cell for CNC suspensions containing PEGDA (P30C5 and P30C6), as presented in Fig. 5. At low shear rates, the suspensions retained a cholesteric arrangement of CNCs at the micrometer scale, leading to the low transmitted intensity observed in Fig. 5a. When shear was applied, the transmitted light intensity increased due to the progressive reduction of cholesteric tactoids. The shear induced a decrease in the size of these cholesteric microdomains, weakening the light structure interaction and reducing scattering, particularly in the shear-thinning regime, where the nuclei were almost in the individual nematic phase.

The normalized dichroism signal $I(\omega)/I_{dc}$ globally increased with the shear rate (Figs. 5b and S7), which is consistent with a growing fraction of CNCs aligning with the flow. Oriented CNCs absorbed polarized light more effectively, which enhanced the dichroic signal. Overall, the dichroism and transmission data illustrated a 3-stage microstructural development under shear: (i) the cholesteric ordering predominated and provided strong scattering and weak dichroism at low shear; (ii) at intermediate shear rate, the cholesteric ordering began to dissociate and the transmitted intensity increased while the dichroic intensity increased as a result of the parallel orientation of the CNCs; (iii) at a higher shear, the CNCs oriented predominantly parallel to the flow and showed an anisotropic but structurally simpler arrangement.

3.2. Structural organization of cross-multilayer CNC films by SAXS/WAXS

3.2.1. Top-view SAXS analysis at the ID02 beamline

The two-dimensional SAXS patterns recorded from the central areas of cross-multilayered CNC nanocomposites, made from P30C6 samples and comprising either 2 or 4 successive layers, are presented in Fig. 6. For samples exhibiting preferential particle alignment, the SAXS pattern became anisotropic, showing an increased scattering intensity in the direction perpendicular to the alignment axis (Fig. S8).

The SAXS analysis was achieved in the top-view orientation, as described in Fig. 2a. As emphasized in Fig. 6a, rectangular scattering patterns were obtained, regardless of the considered number of layers. The observed rectangular orthotropy indicated the coexistence of two orthogonal CNC orientations induced during successive flow-crosslinking steps. This effect was consistently observed for both two- and four-layer constructs, confirming the effectiveness of the fabrication protocol in controlling the reproducibility of CNC alignment within each sheared and crosslinked layer. However, the fact that the observed pattern exhibited a rectangular shape, *i.e.*, a slight departure from the expected square shape, suggested a possible, but slight, difference in CNC preferred alignment between orthogonal layers.

The radially averaged intensity $I(\psi)$ extracted from these rectangular scattering patterns (Fig. 6b for 4 layers) confirmed the persistence of a correlation peak towards ψ = 0, 103, 182, 280, and 360°, corresponding to successive ~90° rotations of CNC orientation in each layer. The

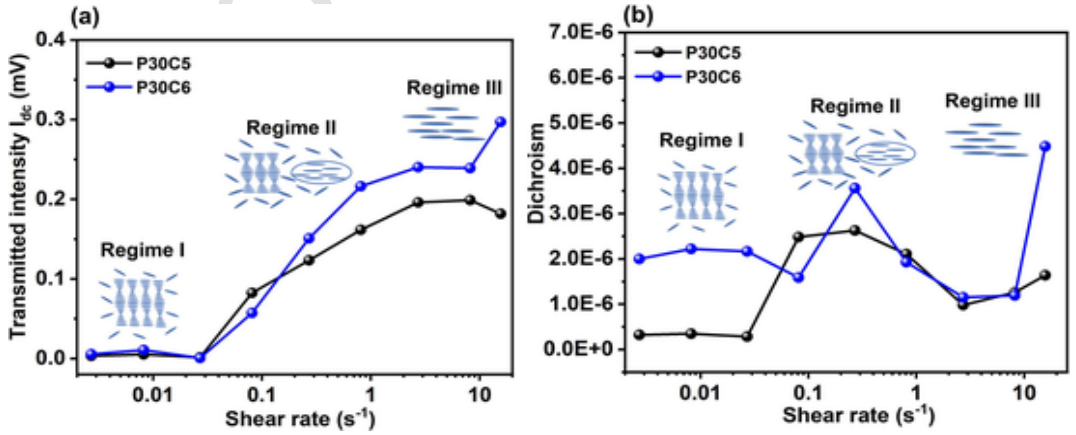


Fig. 5. Comparison of transmitted intensity (a) and dichroism (b) collected from P30C5 and P30C6 at different shear rates, showing regime I associated with the orientation of cholesteric domains, regime II, retaining the tactoids orientation, and regime III, fragmentation of cholesteric domains towards a nematic-like orientation.

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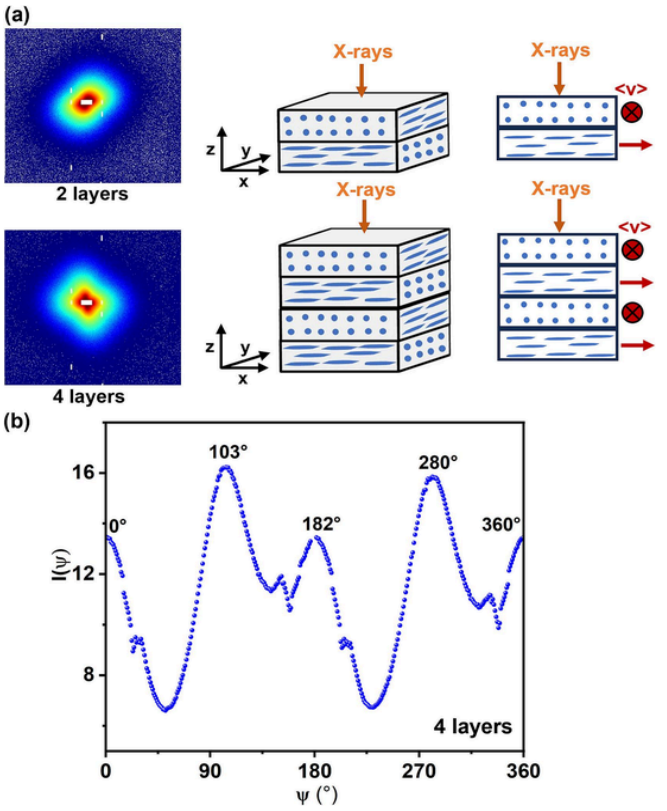


Fig. 6. Normalized 2D SAXS patterns in top view for the P30C6 sample and schematic representation of the SAXS experimental geometry and sample orientation. The *x*, *y*, and *z* axes are defined according to the preferential particle alignment within the sample. The <v> symbol corresponds to the velocity direction associated with the imposed flow during the experiment. (a) Two and 4 layers, and (b) radially averaged intensities characterization along the top view observation of cross-oriented CNC nanocomposites for 4 layers for the P30C6 sample.

higher peak-to-valley modulation in the 4-layer sample showed an enhanced periodicity and long-range order, confirming that the layer-by-layer shear deposition and crosslinking protocol reliably imposed orthogonal orientations. These results, together with the 2D SAXS patterns, provided quantitative evidence of controlled CNC alignment across multilayer architectures.

3.2.2. Side-view SAXS analysis at the ID02 beamline

We also conducted SAXS analysis in the side view observation, as described in Fig. 2b, to characterize the CNC orientation in the different deposited layers independently of one another. In this configuration, a vertical scan through the sample thickness enabled the successive analysis of each layer. For sample P30C5, the normalized 2D SAXS patterns confirmed distinct orientations in both the 2-layer and 4-layer films (Fig. 7a,b). In both cases, a clear cross orientation of CNCs between successive layers was observed, demonstrating the effectiveness of the deposition and crosslinking protocol in producing defined orthogonal arrangements across multilayer nanocomposite architectures. Specifically, the 2D SAXS patterns of the first layer, with slender vertical elliptic shapes, revealed a marked and homogeneous CNC alignment along the midplane layer. In contrast, the second layer exhibited a more isotropic SAXS response in the cross-plane, consistent with reduced orientational order.

The samples with two layers showed uniform thickness homogeneity, with each layer measuring 700 μm. In contrast, the samples with 4 layers exhibited significant variations in thickness: the first layer was 800 μm-thick, the second 600 μm, the third 400 μm, and the fourth 700 μm (Figs. S9 and S10). This indicates that increasing the number of layers can promote a less uniform deposition, resulting in non-homogeneous layer thicknesses across the sample. This behavior can be attributed to variations in material flow, which may cause slight irregularities or partial compression of intermediate layers. Additionally, uneven solidification of earlier layers can affect adhesion and leveling, leading to differences in thickness across the multilayer structure.

The preferred CNC alignment across layers was also quantified and confirmed using the principal component analysis (PCA). As shown in Fig. 7c (two layers), the first layer exhibited the highest degree of orientation, with PCA values ranging from 0.25 to 0.30, consistent with a strong in-plane alignment of CNCs. The addition of a second layer oriented in the cross direction resulted in decreased anisotropy with a PCA value range of 0.01–0.1, which was relative to the overlap of scattering contributions in the perpendicular direction. For the four-layer films (Fig. 7d), values of anisotropy calculated *via* PCA were alternately high and low values following the increasing layers, confirming the successful modulation of the CNC orientation. In particular, the first and third layers showed a higher number of anisotropy values with PCA values ranging from 0.21 to 0.30, consistent with a strong in-plane alignment of CNCs, and the second and fourth layers showed smaller anisotropies, with PCA values ranging from 0.01 to 0.09, predictive of alignment in the perpendicular direction.

These results show clearly and quantitatively that the approach to the multilayer structure through deposition and crosslinking is effective and predictable in inducing alternating perpendicular CNC orientation throughout the entire multilayer structure.

3.2.3. Side and top-view SAXS/WAXS analysis at the ID13 beamline

In order to assess the CNC orientation in the films at the molecular scale, a region of larger *q*-vectors was explored at the ID13 Microfocus beamline. P30C5 and P30C6 with 2 layers composites were raster-scanned using a 1.8 × 2.7 μm^2 X-ray beam and a 5-μm step to scan 5 lines in the samples. The 2D WAXS patterns were indexed according to the crystal structure of the Iβ allomorph of cellulose [54] (Figs. 8 and S11). When P30C5 with 2 layers was analyzed in side view with the X-ray beam perpendicular to the CNCs in layer 1, the fiber pattern exhibited a thin 004 meridional reflection and a strong 2 0 0 equatorial reflection (Fig. 8a,b) [55]. Arcs corresponding to the (1 0 2) and (1 1 0) planes were visible as well. The arcing of the reflections corresponded to a CNC disorientation of about ±16° to the (1 0 2) plane and ± 22° to the (1 1 0) plane. When the beam was parallel to the CNCs in layer 2, the 102 and 004 reflections were absent, consistent with the fact that, along this orientation, the pattern would not contain information along the chain axis (c-axis) of one cellulose crystal (Figs. 8d,e and S11).

Since, in this layer, the CNCs could randomly rotate around their main axis when in suspension, one would have expected the 200 reflections to appear as a continuous ring, like in the idealized pattern in Fig. 8f. However, it is still an arc, albeit wider than that in Fig. 8b, with a disorientation of about ±30°. This effect can be due to the degree of disorientation detected in the longitudinal side view that would also reflect in the axial orientation, but the fact that CNCs can have a preferential orientation around their long axis, which would somewhat induce a textural anisotropy in the scattering pattern, cannot be totally ruled out.

In top view, when the beam crossed both layers, the pattern was the superimposition of two fiber patterns, each corresponding to one layer in which the CNCs were perpendicular to the X-ray beam and thus similar to that in Fig. 8b. Although visible for the P30C5 with 2 layers composite (Fig. 8g,h), the effect was significantly more pronounced in P30C6 with 2 layers, which contained a higher concentration of CNCs (Fig. 8i,j). A series of WAXS patterns recorded from the P30C6 with 4 layers composite showing the alternation of longitudinal and axial pat-

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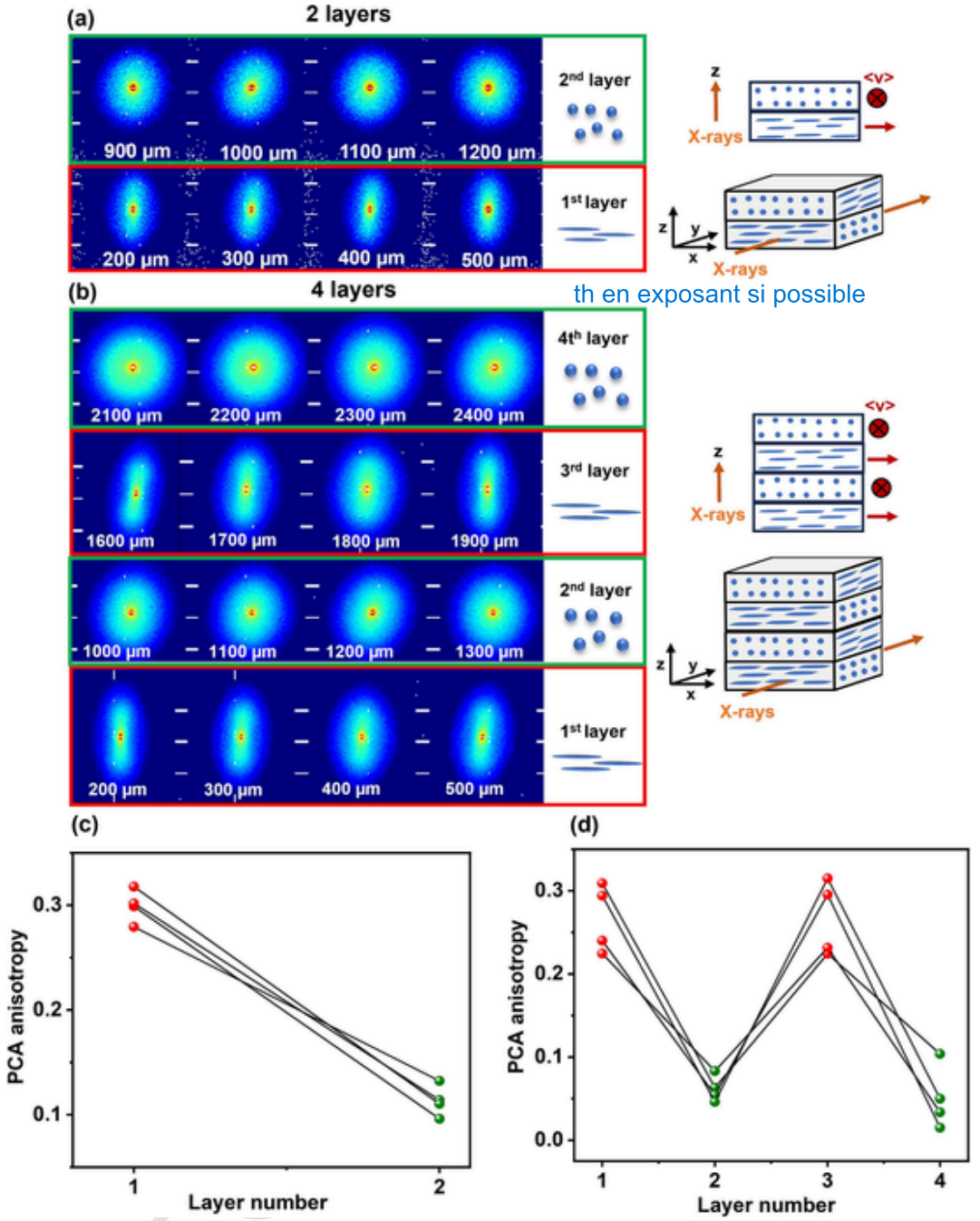


Fig. 7. Normalized 2D SAXS patterns collected at the ID02 beamline in the side view observation for sample P30C5 and schematic representation of the SAXS experimental geometry and sample orientation. The *x*, *y*, and *z* axes are defined according to the preferential particle alignment within the sample. The X-ray beam propagates along the y-direction, while the detector records the scattered intensity in the plane perpendicular to the beam. The symbol <v> denotes the velocity direction associated with the imposed flow during the experiment. Panels show: (a) the patterns obtained for the 2-layer composites, (b) for the 4-layer composites; (c) the PCA anisotropy as a function of the layer number for sample P30C5 with 2 layers, (d) with 4 layers.

terns in side view, and the crossed pattern in top view is also shown in Fig. S12.

The cross orientation of the layers was also reflected by the anisotropy of the scattered intensity in the low-*q* region of the patterns, consistent with the SAXS data collected at the ID02 beamline (Section 3.2.2). The pattern was lozenge for the longitudinal side views, discoidal for the axial side views, and squarish for the top views (Figs. S13 and S14).

A more in-depth analysis of P30C5 with 2 layers was carried out using the annular and sector averages of the scattered intensity. The comparison of the *I(q)* profiles obtained from meridional (260–280°) and equatorial (340–360°) angular sectors showed a difference between the two successive structured layers with CNCs oriented with their main axis along the *x* and *y* axes, respectively (Fig. 9a). When the CNCs were oriented along the *y*-axis, corresponding to an isotropic SAXS pattern, both intensities were similar. When the CNCs were oriented along the *x*-axis, corresponding to anisotropic SAXS patterns, the difference between the two integrations along I_v and I_h was large within the 0.2–2 nm^{-1}, 9–13 nm^{-1}, and 15–17 nm^{-1} regions, confirming the ori-

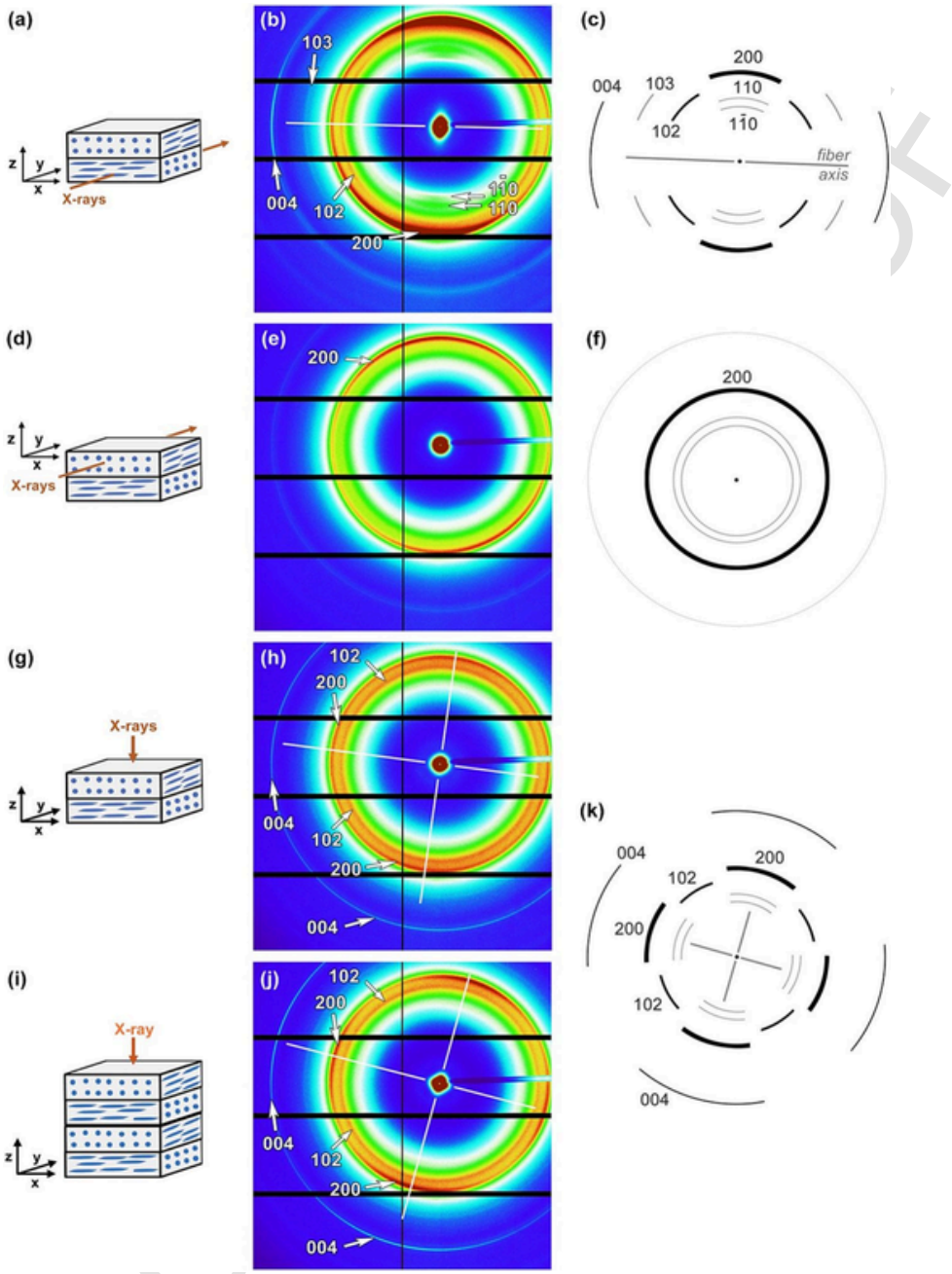


Fig. 8. WAXS patterns of P30C5 with 2 layers (b,e,h) and P30C6 with 2 layers (j) composites recorded along the directions defined in (a), (d), (g), and (i), respectively. The Miller indices of the reflections is indicated in the idealized schemes (c), (f) and (k), respectively, according to the cellulose Iβ allomorph [54] (Fig. S11); (a-c) longitudinal view of layer 1 perpendicular to the CNCs; (d,e) axial view of layer 2 parallel to the long axis of the CNCs; (g,h) top view crossing layers 1 and 2 of P30C5; (i,j) top view crossing layers 1 and 2 of P30C6. The straight lines indicate the average direction of the CNCs with respect to the diffraction patterns.

entation of the CNCs along the *y*-axis with some degree of disorientation.

The annular averages calculated over two *q*-ranges (Fig. S4 c,d) were also calculated and are represented in (Fig. 9b,c) for a set of 20 points separated by 5 μm along the *x* direction, to assess the homogeneity of the structuring and orientation along one plane in each layer. The annular integrations in the SAXS and WAXS range revealed a strong angular dependence of the scattered intensity in layer 2 for the CNCs oriented along the *x*-axis, while a flatter angular profile is detected in layer 1 for the CNCs oriented along the *y*-axis. It is worth noting that the 50 profiles corresponding to 50 positions separated by 5 μm along 250 μm long exhibit the same features, confirming the homogeneity of the orientation of the sample in each layer.

The annular average of the equatorial 200 reflection was calculated over a q-range around 16 nm^{-1} from the WAXS azimuthal representations for both the longitudinal view of layer 1, perpendicular to the CNC alignment (Fig. 9d), and the axial view of layer 2, parallel to the long axis of the CNCs (Fig. 9e). The resulting azimuthal intensity profiles are shown in Fig. 9f. The orientation order parameter, ⟨*f*⟩, and the degree of orientation, Π (Fig. 9g), were determined using Hermans' equations (Eq. (6) and Eq. (9), respectively). Layer 1 exhibited the highest orientation parameter and degree of orientation, indicating a strong in-plane orientation of the CNCs. In contrast, the second layer, de-

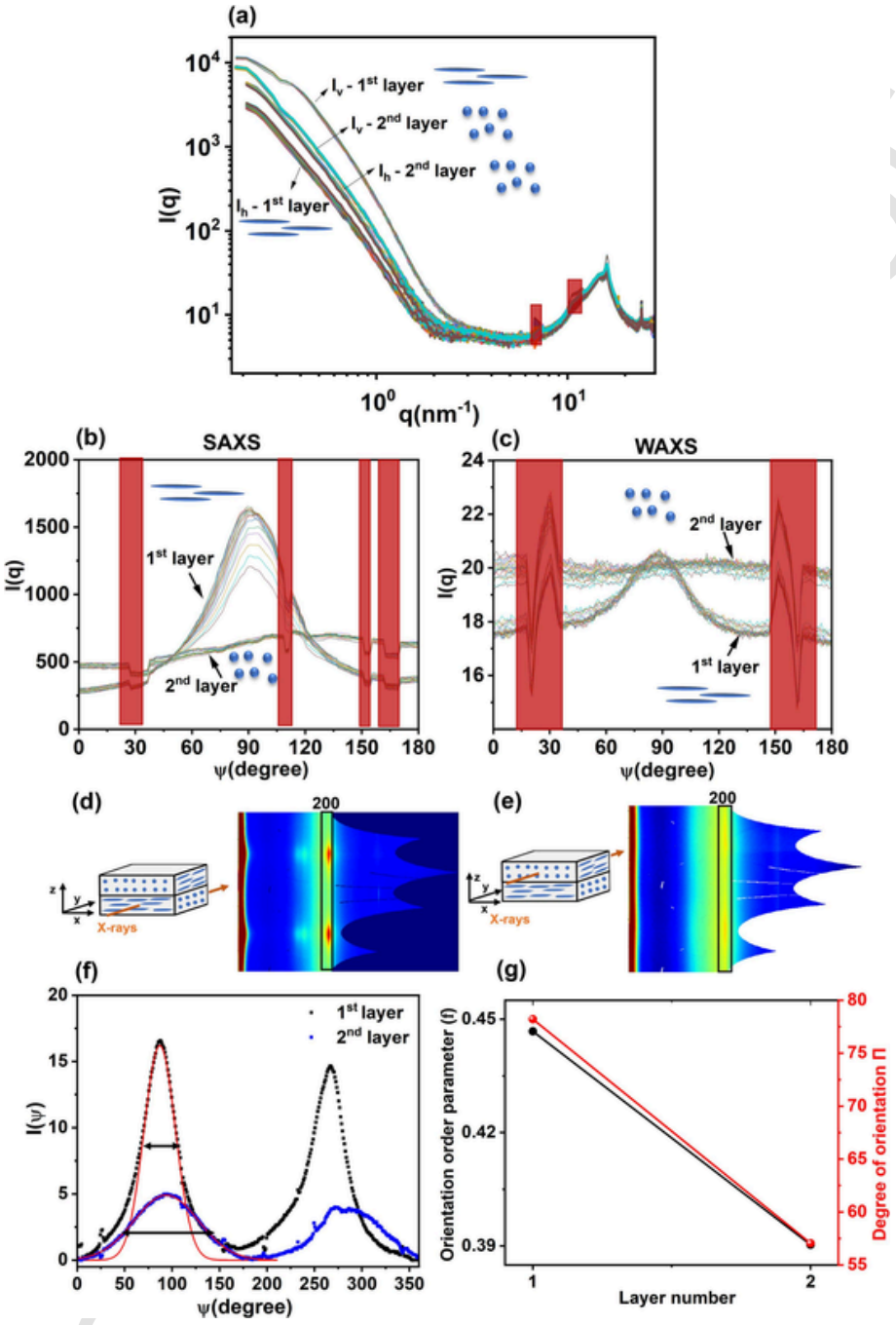


Fig. 9. Annular and sector averages of the scattered intensity for the P30C5 with 2 layers composite: (a) sector angular averaged scattered intensity around meridional (I_v) (260–280°) and equatorial (I_h) (340–360°) angular sectors; (b) annular averaged scattered intensity over a q-range from 0.3 to 1.5 nm^{-1} in the SAXS domain (c) and from 8 to 16 nm^{-1} in the WAXS domain. The red squares indicate the inaccessible region due to the separation between the 8-sensor areas of the 2D EIGER2-4 M detector. 2D-WAXS pattern in azimuthal representation of P30C5 with 2 layers in (d) longitudinal view of layer 1 perpendicular to the CNCs and (e) axial view of layer 2 parallel to the long axis of the CNCs (f) Azimuthal intensity distribution of the equatorial reflection (200) (The red line corresponds to the Gaussian fit and the arrow indicate the FWHM) and (g) orientation order parameter (f) and the degree of orientation, Π as a function of the layer number for sample P30C5 with 2 layers. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

posited in the cross-direction, showed lower orientation, which was relative to the overlap of scattering contributions in the perpendicular direction.

For the top-view analysis of samples P30C6 containing two layers (Fig. S15), the annular average of the 200 plane confirmed the persistence of a correlation peak towards ψ = 73°, 166°, 255°, and 345°, corresponding to successive ~90° rotations of CNC orientation in each layer, associated with enhanced periodicity and long-range structural order, further confirming the successful layer-by-layer deposition process. Similar trends in the annular average profiles, orientation order parameter, and degree of orientation were observed for sample P30C6 with four layers on both side view and top view (Figs. S16 and S17).

Table 1 summarizes the orientation order parameter, *f*, and the degree of orientation, *Π*, for samples studied, deduced from annular-average calculations of the 200 reflections in the WAXS azimuthal patterns. The relationship between the orientation order parameter (*f*) and the degree of orientation (*Π*), Table 1, confirms CNC alignment within the layers in samples P30C5 with 2 layers and P30C6 with 4 layers. Layer 1 and layer 3 present the highest orientation, with *f* values around 0.40–0.44 and *Π* reaching up to 78.2, suggesting a well-organized structure. Layers 2 and 4, exhibit lower values of *f* and *Π*, which was relative to the overlap of scattering contributions in the perpendicular direction.

Table 1
Orientation order parameter, *f* and the degree of orientation *Π* of P30C5 2 layers and P30C6 4 layers samples, deduced from the annular average calculations of the 200 reflections from the WAXS azimuthal patterns.

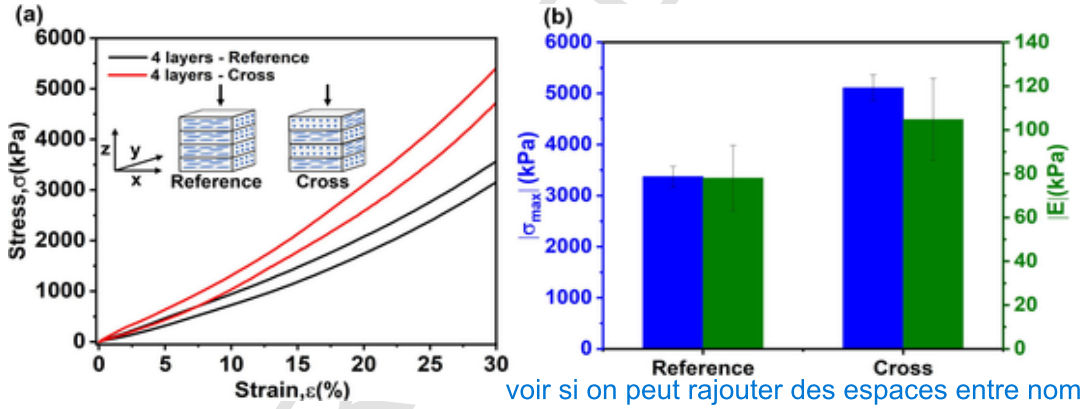
Sample	Layer	*f*	*Π* (%)
P30C5 2L	Layer 1 = =	0.44	78.2
P30C5 2L	Layer 2:::	0.39	57.0
P30C6 4L	Layer 1 = =	0.42	67.1
P30C6 4L	Layer 2::::	0.29	56.1
P30C6 4L	Layer 3 = =	0.40	63.4
P30C6 4L	Layer 4::::	0.21	48.1

3.3. Mechanical properties

Fig. 10 presents the stress-strain responses (Fig. 10a), as well as the maximal stresses, σ_{max}, and the apparent moduli, *E* (Fig. 10b), determined for various 4-layer nanocomposites produced with different stacking configurations when subjected to compression along their thickness, *i.e.*, perpendicular to the CNC orientation. The results are compared for four samples: two reference samples with four identical layers, and two others with layers cross-oriented in pairs. It is interesting to notice that samples with cross-aligned layers exhibit higher stress levels and higher apparent moduli than those reported for uniaxially oriented CNCs (reference). The maximal stress is approximately 51% higher for the cross-aligned samples than for the reference ones, as is the apparent modulus, which is about 34% higher. Indeed, for fully aligned samples, during compression, the in-plane strains mainly occur in the PEGDA matrix transverse to the CNCs, as observed in standard UD fiber-reinforced polymer composites. For cross-aligned samples, this privileged transverse deformation mode is constrained by the perpendicular orientation of CNCs of the neighboring layers, thus demanding higher stress levels.

Different trends are highlighted in Fig. 11, which shows the tensile stress-strain curves (Fig. 11a), strains at break, $\varepsilon_b$as well as the apparent moduli, *E* (Fig. 11b), obtained for the reference and cross-aligned samples, when stretched along (at least one of) the preferred CNC orientations. In contrast to the response in compression, the reference composites (with all CNCs aligned parallel to the loading direction) exhibited

espace dans la légende



voir si on peut rajouter des espaces entre nom et unité dans les légendes x y

Fig. 10. Compression test of the P30C6 films structured in 4 layers with CNCs aligned along the same direction (reference) or with a cross structure (down to $|\varepsilon_{max}| = 30\%$). (a) Stress-strain curves (inset: the *x*, *y*, and *z* axes indicate the preferential particle alignment directions; arrows denote the direction of the applied mechanical load); (b) Maximal stress, σ_{max} and apparent modulus, *E*.

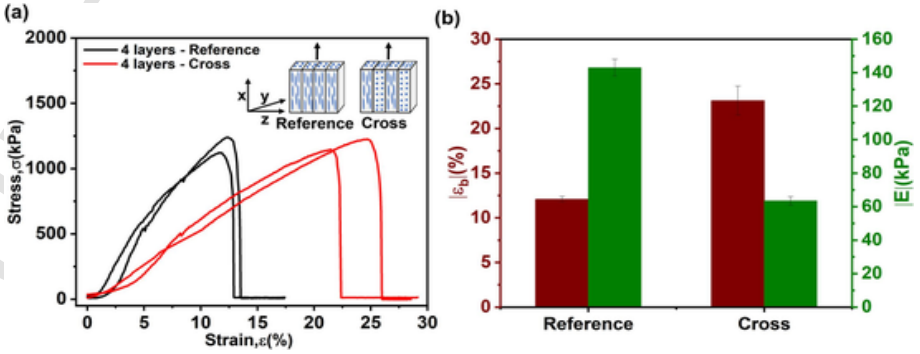


Fig. 11. Tensile test of the P30C6 films structured in 4 layers with CNC aligned in the same direction (reference) or with a cross structure: (a) Stress-strain curves (inset: the *x*, *y*, and *z* axes indicate the preferential particle alignment directions; arrows denote the direction of the applied mechanical load); (b) train at break, $\varepsilon_b$and apparent modulus, *E*.

higher stress levels and apparent moduli (≈125% increase) than the cross-layered samples. The latter, however, were distinguished by a substantially improved ductility (strains at break 91% higher). Note that the ultimate tensile strength (σ_{max}) was comparable for all structures.

Regarding the elastic modulus, this result is in line with the standard behavior of laminated composites: four [0°–0°–0°–0°] layers with fibers aligned along the tensile direction (0°) display higher moduli than four [0°–90°–0°–90°] cross-aligned layers. This behavior comes from efficient load transfer along the fully aligned CNC in the tensile direction. Furthermore, the higher ductility observed in the cross-aligned samples may result from the enhanced deformation of the PEGDA matrix in the 90° layers, as well as by from the progressive rearrangements of the CNCs oriented transversely towards the loading direction. Similar mechanical responses have been reported for shear-aligned CNC films, where tensile modulus strongly depended on the degree of alignment [22,56,57].

Taken together, these results highlight the dual role of orientation and architecture in dictating the performance of CNC-based composites: the cross-layer arrangements developed in this work provide increased stiffness under compression and improved ductility under tension, whereas fully aligned structures offer higher tensile stiffness.

For the uniaxially oriented samples, the tensile properties measured along the orthogonal in-plane direction are significantly lower than those obtained along the CNC alignment direction. This observation confirms a pronounced anisotropic mechanical response attributed to the more efficient load transfer when the applied stress is aligned with the direction of CNC orientation.

In contrast, the alternately oriented (0°/90°) cross-ply samples exhibit comparable tensile properties in both in-plane directions. This behavior suggests a more uniform stress distribution among differently oriented layers, resulting in an overall near-isotropic mechanical response within the plane (Fig. S18).

4. Conclusion

We developed a strategy to fabricate biomimetic nanostructured multilayered CNC-based films with tunable anisotropy, combining shear-induced alignment with controlled layer-by-layer rotational deposition in a PEGDA matrix. SALS and dichroism measurements on CNC suspensions confirmed the onset of flow-induced orientational order, providing the basis for designing stratified architectures. SAXS and WAXS analyses of cross-oriented multilayers validated the nanoscale alignment within each layer and demonstrated the reproducibility of orthogonal CNC orientations across successive depositions. Quantitative anisotropy analysis further revealed the modulation of alignment between layers, confirming the robustness of the deposition protocol. In terms of mechanical behavior, multilayer films containing cross-structured demonstrated improved ductility in tensile deformation and better resistance to compressive deformation than unidirectionally aligned structures. The alternating orientations of the CNCs resulted in improved properties, and naturally stratified composites exhibit similar characteristics as they promote efficient energy dissipation and enhance structural integrity. This study provides a general and scalable means for designing CNC-based nanocomposites with controlled anisotropy. By combining insights from colloidal suspensions with structural and mechanical characterization of solid films, this method proposes new avenues for engineering advanced materials with directional properties for aerospace, automotive, and biomedical applications.

CRediT authorship contribution statement

Elizangela Hafemann Fragal: Writing – review & editing, Writing – original draft, Formal analysis. **Nathan Piaget:** Investigation, Formal analysis. **Mohamed Karrouch:** Methodology, Conceptualization. **Didier Blésès:** Methodology, Conceptualization. **Michael Sztucki:** Formal analysis. **Jean-Luc Putaux:** Writing – review & editing, Resources. **Bruno Jean:** Writing – review & editing, Resources. **Christine Lancelon-Pin:** Formal analysis. **Lucie Bailly:** Writing – review & editing, Resources. **Laurent Orgéas:** Writing – review & editing, Resources. **Michael Holzinger:** Writing – review & editing, Resources. **Nicolas Hengl:** Writing – review & editing, Resources. **Manfred Burghammer:** Formal analysis. **Rémi Tucoulou Tachoueres:** Formal analysis. **Frédéric Pignon:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data are declared in the attach folowing this doi https://doi.org/10.15151/ESRF-ES-2163095176

Scanning SAXS of cellulose nanocrystal composites (Reference data) (ESRF)

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jcis.2026.140776.

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